

Commuting ribbon operators: a classification of the weights

[SKELETON — 17 September 2026. Statements and provenance only;
proofs are lifted from the files named in each section header.]

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Abstract

For $1 \leq e < f$ let R_e^W be the operator on symmetric functions that adds a connected e -ribbon with a weight W depending on the whole shape of the ribbon. We determine every pair $(W, \bar{W})$ with $[R_e^W, R_f^{\bar{W}}] = 0$. Writing $d = \gcd(e, f)$, the answer is a *tensor* statement: the commuting pairs are exactly those obtained from a single rank- d *seed* weight by concatenation, subject to one quadratic condition. The content of that condition is order-theoretic: the possible supports of a seed are the closed sets of a Moore family on $\{0, 1\}^{d-1}$, generated by one local flip rule. Reading the classification off that lattice gives the rest. Its maximal proper elements are $2\lfloor(d-1)/2\rfloor$ explicit “cylinders” of size 2^{d-3} , whence a dichotomy: a nonzero commuting weight either vanishes nowhere or vanishes on at least three quarters of the ribbon shapes of its rank, with nothing in between and both extremes attained. Its one-element closed sets are the *border-split* words, whence, for every $d \geq 3$, operators that add *one ribbon shape and nothing else* and still commute. For $d \leq 2$ the lattice has no proper elements at all, no such operator exists, and the classical Murnaghan–Nakayama sign is forced. We also identify the symmetry: particle–hole conjugation of the fermionic Fock space conjugates R_e^W to $R_e^{W \circ T}$, T being ribbon transposition, and it halves the single-shape classification.

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Status of this file. Skeleton: it fixes the paper's notation once (§2), states the results in full and in their final order (§3), and records for each later section which artifact the proof is lifted from, by filename and label (§4). Nothing here is proved; everything here is *stated* in the paper's notation, which is not the notation of any one source. §7 is the honest list of what remains open. §5 records the four points at which a source is not quoted verbatim; as of 18 September 2026 all four are *applied* in the text above it, and that section is a reader's note, not a to-do list.

1 Introduction

To be written last. The arc: Murnaghan–Nakayama says that multiplying by a power sum adds ribbons with the sign $(-1)^{\text{ht}}$, and that these operators commute. Ask which *other* weights make them commute. If the weight may see only the height of the ribbon, the answer is: none but Murnaghan–Nakayama (§4). If it may see the whole shape, the answer is strictly larger as soon as $\gcd(e, f) \geq 3$, and the extra solutions are not deformations of Murnaghan–Nakayama — they are operators with large zero sets, the extreme case adding a single ribbon shape (§4). The organising fact is that the one-bead sector of the commutator is literally the statement that two words commute in a free monoid, so the gcd in the answer is the gcd of Lyndon–Schützenberger.

2 Notation, fixed once

Convention 2.1 (the objects). K is an integral domain with fraction field F . $\Lambda = \Lambda_K$ is the ring of symmetric functions with Schur basis $\{s_\lambda\}$ and Hall inner product. For a partition λ , $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$ is its *Maya set*; elements are *beads*, elements of $\mathbb{Z} \setminus M$ *holes*. $\mathcal{F}$ is the charge-0 semi-infinite wedge with basis $|M\rangle$, identified with Λ by $|M(\lambda)\rangle \leftrightarrow s_\lambda$; the free fermions are undeformed, $\{\psi_a, \psi_b^*\} = \delta_{ab}$, and $n_a = \psi_a \psi_a^*$.

Convention 2.2 (words, shapes, weights). A legal *rank- e move* in M is $b \mapsto b + e$ with $b \in M$, $b + e \notin M$; its *occupancy word* is $u \in \{0, 1\}^{e-1}$, $u_i = [b + i \in M]$. We write $|u| = \sum u_i$ and $\text{ht} = |u|$, the number of rows of the ribbon minus one. A *shape weight of rank e* is a function $W : \{0, 1\}^{e-1} \rightarrow K$; it defines

$$R_e^W s_\lambda = \sum_{\mu} W(u) s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected e -ribbon of occupancy word u . The *sign-normalised* weight is $\sigma(u) := (-1)^{|u|} W(u)$; Murnaghan–Nakayama is $\sigma \equiv \text{const}$, i.e. $R_e^W = \sigma_0(p_e \cdot)$. A *height weight* is one with $W(u)$ depending only on $|u|$.

Convention 2.3 (the three involutions of $\{0, 1\}^n$). $(\text{rev } u)_i = u_{n+1-i}$, $(\text{cpl } u)_i = 1 - u_i$, and $T := \text{cpl rev} = \text{rev cpl}$, so $(u^T)_i = 1 - u_{n+1-i}$. T is ribbon transposition. $W^\vee := W \circ T$.

Convention 2.4 (concatenation). For shape weights σ_1, σ_2 of ranks m_1, m_2 ,

$$(\sigma_1 \otimes \sigma_2)(y) := \sigma_1(y_{[1, m_1-1]}) \sigma_2(y_{[m_1+1, m_1+m_2-1]}), \quad y \in \{0, 1\}^{m_1+m_2-1},$$

the letter y_{m_1} being ignored; $\otimes$ is associative, and $\sigma^{\otimes k}$ is the k -fold power.

Convention 2.5 (the two quadratic conditions). A rank- m shape weight σ satisfies $(2\mathbf{B}_m^*)$ if for all $\delta \in [1, m-1]$, $\pi, \varsigma \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{m-\delta-1}$,

$$\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) = \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma);$$

and $(2\mathbf{B}_m^{**})$ if the same holds with the second factor's prefix decoupled. $(2\mathbf{B}^{**}) \Rightarrow (2\mathbf{B}^*)$, strictly from $m \geq 4$.

Convention 2.6 (border-split words). Let $w \in \{0, 1\}^n$. An integer $p \in [0, n-1]$ is a *border length* of w if $w_{[1,p]} = w_{[n-p+1,n]}$ (so $p = 0$ always is one). Call w **border-split** if

$$w_{p+1} \neq w_{n-p} \quad \text{for every border length } p \text{ of } w. \quad (\text{BS})$$

Remark 2.7 (why not “non-overlapping”). [6] calls (BS) “non-overlapping”. That term is already taken: in combinatorics on words a non-overlapping (= unbordered, bifix-free) word is one with no border length $p \in [1, n-1]$ at all. (BS) is strictly weaker — it permits a border provided the two continuation letters differ. The families differ from $n = 4$ ($d = 5$) on: 8 border-split words against 6 unbordered ones, the difference being 0101 and 1010. The counts 0, 2, 4, 8, 12, 28, 48, 104 for $d = 2, \dots, 9$ reported in [6, §9] are the (BS) counts, and are correct; only the name is hazardous. **This paper imports the condition and not the word.**

2.1 Reconciliation with the five sources

The five artifacts do not use one convention. The table records every symbol whose meaning changes between them. A row marked $\blacklozenge$ is a genuine collision — the same symbol naming two different objects — and is the reason this section exists.

symbol	Q91 / Q92 / Q147	Q149	Q150c1 / Q150c2	this paper
$\blacklozenge \setminus \mathbf{F}$	$\mathcal{F}$, the Fock space	—	$\mathbb{F}$, a <i>field</i>	$\setminus \mathbf{Fock} = \mathcal{F}$ (space), $\setminus \mathbf{Fld} = F$ (field)
$\blacklozenge \mathcal{C}$	—	—	$\mathcal{C}_{j,b}$, cylinder sets ($\setminus \mathbf{Cset}$)	$\mathcal{C}_{j,b}$ cylinders; particle-hole conjugation is $\mathcal{P}$, <i>never</i> $\mathcal{C}$
$\blacklozenge \tau$	Q147: τ_N , the height weight in fermionic coordinates, $W_N = (-1)^N \tau_N$	—	the rank- d seed weight of the tensor decomposition	τ = seed only. Q147's τ_N is eliminated: we state Q147 in W/σ coordinates throughout
$\blacklozenge w$	Q147: w_N , a height weight	—	Q150c2: $w \in \{0, 1\}^n$, a <i>word</i>	w = word only; height weights are W with $W(u) = W_{ u }$
$\blacklozenge \gamma$	—	$\gamma : \mathbb{Z}/d \rightarrow F^\times$, a character	Q150c1/c2 proofs: $\gamma \in \{0, 1\}$, a single <i>letter</i>	γ = character; letters are b
$\blacklozenge \alpha$	—	$\alpha(u)$, the composition (§1); <i>and</i> $\alpha \in K^\times$, a scalar (§5)	$\alpha(w)$, the composition	$\alpha(u)$ = composition; scalars are $c, \bar{c}$

symbol	Q91 / Q92 / Q147	Q149	Q150c1 / Q150c2	this paper
♦ k	Q92: $k_{b,c}^{e,f}$, the interference index	—	Q150c2: k , the tensor power	k = tensor power; the interference index is $\varkappa_{b,c}^{e,f}$
♦ m	$m(x) = [x \in M]$; Q147 $m_e = m(b+e)$	—	m_1, m_2 , ranks	$m(x)$ = bead indicator; ranks are e, f, d
♦ N	$N_{(x,y)} = \sum_{x < j < y} n_j$, an operator; Q147 N = a height index	—	—	$N_{(x,y)}$ operator only; height indices are h
<code>\Mult{p_e}</code>	M_{p_e}	$p_e \cdot$	$p_e \cdot$	$p_e \cdot$ (macro dropped)
height macro	<code>\hgt</code>	<code>\hggt</code>	<code>\height</code> (c1), <code>\hgt</code> (c2)	<code>\hgt</code> , one macro
the operator	$R_e(t)$ (weight t^{ht} built in)	R_e^W	R_e^W	R_e^W ; $R_e(t) := R_e^W$ with $W(u) = t^{ u }$
σ	—	$\sigma(u)$ $(-1)^{ u }W(u)$	= same; $\sigma, \bar{\sigma}$ of ranks e, f	same
d	—	$d = \gcd(e, f)$	= $\gcd(e, f)$ (c1); the rank of a single weight (c2)	$d = \gcd(e, f)$ always; a bare rank is m
n	—	—	Q150c2: $n = d - 1$, the word length	n = word length, stated each time
(NB)	—	—	Q150c2 redefines “non-overlapping”	(BS), border-split (Rem. 2.7)

Remark 2.8 (one sign convention, stated once). Q147 records $W_N = (-1)^N \tau_N$: the $(-1)^N$ inside $(-t)^N$ in the fermionic normal form is not part of the weight, it cancels the Jordan–Wigner sign. We never write τ_N , so this identity is folded into Convention 2.2 and never reappears. The one place the sign is visible is $\sigma = (-1)^{|u|}W$: in σ -coordinates every matrix element of the commutator is a difference of two products with $+$ signs.

3 The results

The classification (Theorem 3.11) reduces the whole commuting locus to a single object: one shape weight τ of rank $d = \gcd(e, f)$, subject to one quadratic condition (2B*). Everything else — the dichotomy, the single-shape operators, the counts — is a statement about the solutions of that condition, and those solutions are governed by an order-theoretic structure. We therefore state the order theory first (§3.1) and feed it back into the operator statements afterwards (§3.2). Throughout, $n := d - 1$ is the word length.

3.1 The admissible sets form a closure system

Definition 3.1 (admissible sets). $Z \subseteq \{0, 1\}^n$ is *admissible* (for rank $d = n + 1$) if for every $\delta \in [1, d - 1]$ and all $\pi, \varsigma \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{d-\delta-1}$,

$$[\pi 0 \pi' \in Z \wedge \pi' 0 \varsigma \in Z] \iff [\pi 1 \pi' \in Z \wedge \pi' 1 \varsigma \in Z].$$

Proposition 3.2 (admissible = support of a solution). *Let σ be a rank- d shape weight over the domain K and $Z = \text{supp}(\sigma)$. If σ satisfies $(2B_d^*)$ then Z is admissible; conversely if Z is admissible then $\mathbb{K}_Z$ satisfies $(2B_d^*)$. Hence the admissible subsets of $\{0, 1\}^n$ are exactly the supports of $(2B_d^*)$ -solutions.*

So the combinatorics of the commuting locus is the combinatorics of the family of admissible sets. That family is not merely a list: it is closed under intersection, and it is generated by a single local rule.

Definition 3.3 (linked pairs and the flip). For $u, v \in \{0, 1\}^n$ and $\delta \in [1, n]$, the pair (u, v) is δ -linked if

$$u_{[\delta+1, n]} = v_{[1, n-\delta]} \quad \text{and} \quad u_\delta = v_{n-\delta+1}.$$

Writing $x^{(i)}$ for x with the letter at position i flipped, set $\iota_\delta(u, v) := (u^{(\delta)}, v^{(n-\delta+1)})$; this is an involution of the set of δ -linked pairs.

Theorem 3.4 (generation). $Z \subseteq \{0, 1\}^n$ is admissible if and only if it is closed under the flip rule

$$u, v \in Z \text{ } \delta\text{-linked} \implies u^{(\delta)} \in Z \text{ and } v^{(n-\delta+1)} \in Z \quad (1 \leq \delta \leq n). \quad (\text{R})$$

Proposition 3.5 (the Moore family). *The admissible subsets of $\{0, 1\}^n$ are closed under arbitrary intersections, and $\{0, 1\}^n$ is itself admissible. They therefore form a closure system: there is a closure operator $X \mapsto \langle X \rangle$ sending a set of words to the least admissible set containing it, and the admissible sets are exactly the sets $\langle X \rangle$.*

This is the result the paper is organised around. A biconditional condition (Definition 3.1) has no reason to define a closure system — intersections of solution sets of biconditionals are not solution sets in general — and it is Theorem 3.4, which replaces the biconditional by an implication, that makes it one. Everything below is obtained by locating specific elements of this lattice: its maximal proper elements (Corollary 3.8), which give the dichotomy, and its atoms (Corollary 3.10), which give the single-shape operators.

Definition 3.6 (cylinders). For $1 \leq j \leq d-1$ with $2j \neq d$ and $b \in \{0, 1\}$ put $\mathcal{C}_{j,b} := \{w \in \{0, 1\}^n : w_j = b, w_{d-j} = 1-b\}$, so $|\mathcal{C}_{j,b}| = 2^{d-3}$. Since $\mathcal{C}_{j,b} = \mathcal{C}_{d-j,1-b}$ there are exactly $2\lfloor (d-1)/2 \rfloor$ distinct cylinders, and none at all for $d \leq 2$.

Theorem 3.7 (structure of an admissible set). *Let $d \geq 2$ and let $Z \subseteq \{0, 1\}^n$ be admissible. Then $Z = \emptyset$, or $Z = \{0, 1\}^n$, or $Z \subseteq \mathcal{C}_{j,b}$ for some $j \in [1, d-1]$ with $2j \neq d$ and some $b \in \{0, 1\}$. In the third case, if j_0 is the least such j , then Z is invariant under flipping any single coordinate in $\{1, \dots, j_0-1\} \cup \{d-j_0+1, \dots, d-1\}$.*

Corollary 3.8 (the maximal proper admissible sets are exactly the cylinders). *Let $d \geq 3$. The maximal proper admissible subsets of $\{0, 1\}^n$ are precisely the $2\lfloor (d-1)/2 \rfloor$ distinct cylinders $\mathcal{C}_{j,b}$. For $d \leq 2$ there is no proper nonempty admissible set at all.*

Counting the maximal elements now gives the density statement, as a corollary rather than a theorem.

Corollary 3.9 (the quarter dichotomy). *Let $\tau \neq 0$ be a rank- d shape weight over K satisfying $(2B_d^*)$. Then either τ vanishes nowhere, or*

$$\#\text{supp}(\tau) \leq 2^{d-3} = \frac{1}{4} \cdot 2^{d-1},$$

i.e. τ vanishes on at least three quarters of the 2^{d-1} ribbon shapes of rank d . There is nothing in between. Both alternatives occur, and the bound 2^{d-3} is attained, by $\mathbb{K}_{c,j,b}$ for any admissible index (j,b) ; so “at least” is sharp and cannot be improved to “more than”. For $d \leq 2$ only the first alternative occurs.

Corollary 3.10 (the self-closed words). *For $w \in \{0,1\}^n$: $\langle\{w\}\rangle = \{w\}$ if and only if w is border-split (Convention 2.6). Equivalently, the singletons among the atoms of the closure system are exactly the border-split words; their number is 2, 4, 8, 12, 28, 48, 104 for $d = 3, \dots, 9$.*

Indeed rule (R) applied to $u = v = w$ is nontrivial exactly when (w, w) is δ -linked, i.e. when w has period δ and $w_\delta = w_{n-\delta+1}$ — which is the negation of (BS). Corollary 3.10 is what makes Theorem 3.12 below a statement about this lattice rather than a separate computation.

3.2 The operator statements

Theorem 3.11 (classification). *Let $1 \leq e < f$, $d = \gcd(e, f)$, and let $W, \bar{W}$ be shape weights of ranks e, f over K with $W \neq 0 \neq \bar{W}$. Then $[R_e^W, R_f^{\bar{W}}] = 0$ if and only if there are a nonzero shape weight τ of rank d over F and $c, \bar{c} \in F^\times$ with*

$$\sigma = c\tau^{\otimes(e/d)}, \quad \bar{\sigma} = \bar{c}\tau^{\otimes(f/d)},$$

and σ satisfies $(2B_e^*)$ (vacuous for $e = 1$).

Theorem 3.12 (single-shape commuting operators). *Let $d \geq 3$, let $w \in \{0,1\}^{d-1}$ be border-split, let $\alpha(w) \models d$ be the corresponding ribbon shape, let $k \geq 2$ and put $e = d$, $f = kd$. Let $W = \mathbb{K}_w$ and let $\bar{W}(y) = (-1)^{\text{ht}(y)} \prod_{i=1}^k [y_{[(i-1)d+1, id-1]} = w]$. Then R_d^W adds the single ribbon shape $\alpha(w)$ and nothing else, $\bar{W}$ is supported on exactly 2^{k-1} of the 2^{kd-1} shapes of rank kd , and $[R_d^W, R_{kd}^{\bar{W}}] = 0$. The number of admissible w is 2, 4, 8, 12, 28, 48, 104 for $d = 3, \dots, 9$.*

By Corollary 3.10 these are exactly the operators whose support is a *one-element* admissible set; by Corollary 3.8 the cylinder indicators are the operators whose support is a *maximal proper* one. The two extremes of the same lattice give, respectively, the sparsest commuting operators and the densest vanishing ones.

Theorem 3.13 (the symmetry). *Let $\mathcal{P}(M) = \{-1 - x : x \in \mathbb{Z} \setminus M\}$ be particle-hole conjugation, an involution of charge-0 Maya sets realising $s_\lambda \mapsto s_{\lambda'}$. Then for every e and W ,*

$$\mathcal{P} R_e^W \mathcal{P} = R_e^{W \circ T},$$

so $X \mapsto \mathcal{P}X\mathcal{P}$ is an algebra automorphism preserving the commuting locus. On the nowhere-zero part of Theorem 3.11 it acts by a scalar; on the single-shape part it acts by the involution $w \mapsto w^T$ of border-split words, whose fixed points are exactly the antipalindromes, i.e. exactly the w for which $\mathbb{K}_w$ satisfies $(2B_d^{**})$. Hence the single-shape classification may be carried out on T -orbits: 2, 2, 6, 6, 18, 24, 60 orbits for $d = 3, \dots, 9$, against 2, 4, 8, 12, 28, 48, 104 words and 2, 0, 4, 0, 8, 0, 16 antipalindromes.

Remark 3.14 (a corrected count). [7, V10] reports the orbit counts as 1, 2, 2, 6, 6, 18, 24, 60 for $d = 2, \dots, 9$. The leading 1 is wrong and is corrected here to 0, i.e. the list is reindexed to start at $d = 3$. There are no border-split words of length $n = 1$ at all — the empty border $p = 0$ forces $w_1 \neq w_1$ — so the same source’s own shape counts begin 0, 2, 4, $\dots$ at $d = 2$, and an empty set has no orbits. The remaining seven terms are correct: by Burnside the orbit count is $\frac{1}{2}(\#\text{words} + \#\text{antipalindromes})$, and $(2+2)/2, (4+0)/2, (8+4)/2, (12+0)/2, (28+8)/2, (48+0)/2, (104+16)/2 = 2, 2, 6, 6, 18, 24, 60$ as stated. The error is confined to a value of d at which Theorem 3.12 does not apply, so nothing downstream of it changes.

Corollary 3.15 (the coprime headline). *If $\gcd(e, f) = 1$ then τ has rank 1, so σ and $\bar{\sigma}$ are constant: $R_e^W = c(p_{e\cdot})$ and $R_f^{\bar{W}} = \bar{c}(p_{f\cdot})$. Shape dependence is forced away, and with it every parameter but one per rank.*

Corollary 3.16 (height weights). *Restricted to height weights the classification collapses for all $e < f$: $[R_e^W, R_f^{\bar{W}}] = 0$ with both nonzero forces $W(u) = (-1)^{|u|}W_0$ and $\bar{W}(y) = (-1)^{|y|}\bar{W}_0$.*

Remark 3.17 (the threshold, conceptually). Definition 3.6 is empty for $d \leq 2$ and nonempty for $d \geq 3$, because $d \geq 3$ is exactly the condition that $[1, d-1]$ contains an index j with $j \neq d-j$. That single fact is the whole of the threshold $\gcd(e, f) \geq 3$: below it Theorem 3.7 leaves only $\emptyset$ and $\{0, 1\}^n$, so every commuting weight is nowhere zero and Corollary 3.15's conclusion propagates; at and above it the lattice acquires proper elements, and with them the operators of Theorem 3.12.

4 Section skeleton and provenance

Each subsection below states what the corresponding section of the paper proves and the file and label the proof is lifted from. Numbers are given from the label map of Appendix A, generated from the .aux files on 17 September 2026.

§4 The abacus dictionary and the fermionic normal form

Proves: adding an e -ribbon is a bead move $b \mapsto b + e$; the height is the number of beads jumped; the occupancy word determines the ribbon shape, *top row first*, by $\alpha(u) = (c_{k+1} - c_k, \dots, c_1 - c_0)$ where the c_i list $\{0\} \cup \{i : u_i = 1\} \cup \{e\}$; and $R_e(t) = \sum_b \psi_{b+e} \psi_b^* (-t)^{N(b, b+e)}$. *Lifted from:* 2026-09-06-c2-Q91-fermionic-normal-form.tex, lem:abacus, lem:sign, lem:dressing, thm:main (=Thm. 7); and 2026-09-15-c1-Q149-shape-weights.tex, lem:dict (=Lem. 1.3). *Note:* the reversal in $\alpha(u)$ is not cosmetic — without it the two L-trominoes exchange names, which is exactly the distinction the cheapest test of the paper turns on.

§5 The two sectors of the commutator

Proves: $\langle s_\mu, [R_e^W, R_f^{\bar{W}}] s_\lambda \rangle = 0$ unless $|M(\lambda) \triangle M(\mu)| \in \{2, 4\}$; closed forms for the one-bead and two-bead matrix elements in σ -coordinates. *Lifted from:* 2026-09-15-c1-Q149-shape-weights.tex, lem:sectors (=Lem. 4.1), lem:onebead (=Lem. 4.2), lem:twobead (=Lem. 4.3). *Background, for the single- t case only:* 2026-09-07-Q92-cross-rank-commutator.tex, thm:matrix (=Thm. 5), giving $(m(b+e) - m(b+f))(1+t)t^{N-1}$ in the one-bead sector and $t^{P+Q}(t^{-\varkappa} - t^\varkappa)$ in the two-bead sector, and thm:main (=Thm. 6), $[R_e(t), R_f(t)] = -\frac{1+t}{t}(\Phi_{e,f} + (t-1)\Psi_{e,f})$. *Note:* the witness $\langle s_\mu, [R_e(t), R_f(t)] s_\lambda \rangle = t^f - t^{f-2}$ is prop:psiwitness (=Prop. 16) and belongs to the **two**-bead sector; the brief for this session filed it under the one-bead sector.

§6 Height weights force Murnaghan–Nakayama

Proves: Corollary 3.16, using only the one-bead sector and only the occupancy pattern $m_e = 1$, $m_f = 0$. *Lifted from:* 2026-09-11-c2-Q147-multi-t-bracket.tex, thm:main (=Thm. 7) via lem:B, with lem:realise as the control that every window configuration occurs. *Note:* restate in W -coordinates; do not import τ_N (Remark 2.8). The $(1+t)$ factor of Q92 is the $N = 0$ instance of the forcing relation $W_{N+1} = -W_N$.

§7 The one-bead sector is commutation in a free monoid

Proves: the one-bead sector vanishes identically iff $\bar{\sigma} = \sigma \otimes \rho = \rho \otimes \sigma$ for a unique nonzero ρ of rank $f - e$; and two shape weights commute under $\otimes$ iff both are powers of a common seed — Lyndon–Schützenberger for the semigroup of Convention 2.4, by Euclidean descent. This is where the gcd comes from. *Lifted from:* 2026-09-16-c1-Q150-nonvanishing.tex, prop:1B (=Prop. 3.1) and thm:structure0 (=Thm. 1.2).

§8 The two-bead sector is $(2B^*)$, and the classification

Proves: given the one-bead conclusion, the two-bead sector is exactly $(2B_e^*)$ for σ ; hence Theorem 3.11. *Lifted from:* 2026-09-16-c1-Q150-nonvanishing.tex, prop:2B and thm:main0. *Note:* thm:main0 is a biconditional, proved from the two biconditionals prop:1B, prop:2B and [4, lem:sectors] alone. In particular **both** directions — including sufficiency, for which [4, prop:suff] would be the alternative source — are available without any appeal to nonvanishing. This is why §5(P2) is an absence of dependency rather than a repair. *Note:* prop:2B’s converse direction makes a *second, separate* appeal to K being a domain, with witness $\rho(\psi_0)$; say so, per [10, one-domain-appeal].

§9 The structure of an admissible support, and the dichotomy

Proves: an admissible support is empty, everything, or contained in a cylinder $\mathcal{C}_{j,b}$ with $2j \neq d$; the maximal proper admissible sets are exactly the $2\lfloor(d-1)/2\rfloor$ cylinders; hence Theorem 3.7, Corollary 3.8 and, by counting them, Corollary 3.9. Also Remark 3.17: $d \geq 3$ is exactly the condition that $[1, d-1]$ contains some $j \neq d-j$, which is the conceptual content of the threshold $\gcd(e, f) \geq 3$. *Lifted from:* 2026-09-16-c2-Q150-H2-single-shape.tex, def:S (=Def. 2.2), prop:supp (=Prop. 2.3), thm:struct (=Thm. 6.1), cor:three (=Cor. 6.2), cor:maximal (=Cor. 6.3), cor:quarter (=Cor. 6.4). *Correction applied:* the source’s prose gloss says “more than three quarters”; Corollary 3.9 says “at least”, which is what its own display asserts and what the cylinder indicator attains. See §5(P3).

§10 Single-shape weights

Proves: $\mathbb{K}_w$ satisfies $(2B_d^*)$ iff w is border-split; it satisfies $(2B_d^{**})$ iff w is an antipalindrome, of which there are $2^{(d-1)/2}$ for d odd and none for d even; hence Theorem 3.12, with the smallest new case $d = 4, k = 2, w = 001$, shape $(1, 3)$. *Lifted from:* 2026-09-16-c2-Q150-H2-single-shape.tex, def:border (=Def. 4.1), thm:singleton (=Thm. 4.2), rem:endrows (=Rem. 4.3), thm:singleton2 (=Thm. 4.5), thm:main (=Thm. 5.1), cor:48 (=Cor. 5.2). *Correction applied:* the source’s “non-overlapping” is imported as the condition (BS) and not as the word; see Convention 2.6, Remark 2.7 and §5(P1). *Note:* for even d these operators are new precisely because the $(2B^{**})$ route produces none of them — for d even the word length $n = d - 1$ is *odd*, and an antipalindrome $w = w^T$ of odd length would need $w_{(n+1)/2} = 1 - w_{(n+1)/2}$ at its middle letter. (The paper’s §7(H1) states this correctly; an earlier draft of this note said “even length”, which is the same fact with the parity of n and of d confused.)

§11 Generation: the admissible sets form a closure system

Proves: Definition 3.3, Theorem 3.4 and Proposition 3.5 — Z is admissible iff closed under the flip rule, so the admissible sets are the closed sets of a Moore family — together with Corollary 3.10, $\langle\{w\}\rangle = \{w\}$ iff w is border-split. *Lifted from:* 2026-09-16-c2-Q150-H2-single-shape.tex,

`def:linked`, `thm:closure`, `prop:moore`, and for Corollary 3.10 the remark immediately following `prop:moore` there, read together with `thm:singleton`. *Note on placement:* in the source this is the *last* section, stated as an afterthought to the single-shape classification. It is the first result here. The proofs are unchanged; only their order is. Proposition 3.5’s proof is three lines and uses nothing but Theorem 3.4, so nothing is lost by promoting it.

§12 Particle–hole conjugation

Proves: Theorem 3.13; that $\mathcal{P}$ -conjugation is an *automorphism*, not an antiautomorphism, so it is not Terwilliger’s $\dagger$; and that no invertible graded linear map implements `rev` or `cpl` alone, so T is exactly the part of the word-level $(\mathbb{Z}/2)^2$ that an operator can see. *Lifted from:* 2026-09-17-c1-Q151-particle-hole
`thm:A`, `lem:Ctrans`, `thm:C`, `thm:D`, `cor:parity`, `prop:norev`, `cor:accounting`. *Note:* `thm:A` also upgrades `[4, lem:transpose]` (=Lem. 1.6) from `computed` to `proved`: the fact that transposition is reverse-and-complement on occupancy words, previously checked on 1500 ribbons, has a three-line proof on the bead lattice.

5 Provenance note: four places where a source is not quoted verbatim

The corrections below have been *applied* in §2 and §3; this section exists so that a reader holding the sources can see where this paper and they part company, and why. None is a gap in a proof.

- (P1) **The word “non-overlapping” is not imported.** `[6, def:border]` uses it for the condition called (BS) in Convention 2.6. In combinatorics on words a non-overlapping (= unbordered, bifix-free) word is one with no border length $p \in [1, n - 1]$ at all; (BS) is strictly weaker, permitting a border provided the two continuation letters differ. The families first differ at $n = 4$ ($d = 5$): 8 border-split words against 6 unbordered ones, the difference being 0101 and 1010. The counts 0, 2, 4, 8, 12, 28, 48, 104 for $d = 2, \dots, 9$ reported there are the (BS) counts and are correct; only the name is hazardous. (Remark 2.7.)
- (P2) **Nothing is quoted from [4] §5, §7 or §8.** This is the one item with correctness at stake, and the resolution is stronger than a repair: the dependency is *absent*, not corrected. `[4, lem:nonvanish]` (“a nonzero commuting weight vanishes nowhere”) is **false** for $\gcd(e, f) \geq 3$ `[5, cor:main0]`, and `[4]` itself flags it as “the one gap of the paper” for $e \geq 3$, adding “everything after this point assumes it”. That warning is exact. It does not reach this paper, because every statement of §3.2 is taken from `[5, thm:main0]`, which is a biconditional proved from `[5, prop:1B]` and `[5, prop:2B]` — both biconditionals — together with `[4, lem:sectors]`, and *uses no nonvanishing input in either direction*. The citations of `[4]` surviving in §4 are `lem:dict`, `lem:transpose`, `lem:realise`, `lem:sectors`, `lem:onebead`, `lem:twobead` — all in §§1,3,4 of that paper, all unconditional, none touched by the audit.

The audit is retained below because `[4]` remains on the shelf and will be cited by later work; it records how far the flaw reaches, which the blanket instruction “re-check all of `[4, §8]`” is too coarse to say. Two of its five results are unconditional, and two of those — `lem:flip` and `prop:e1` — are in any case superseded by `[5, prop:1B]` and reproved by `[5, cor:coprime]` respectively, so even a reader who wanted them need not go back.

label	§	status
lem:flip (F0)–(F3)	8	Unconditional. Its proof needs only $\sigma \neq 0 \neq \bar{\sigma}$ and K a domain — it picks <i>some</i> nonzero witness $\sigma(A)$ or $\sigma(D)$, never all of them. Superseded by [5, prop:1B].
lem:nonvanish	8	False for $\gcd(e, f) \geq 3$. True exactly for $\gcd(e, f) \leq 2$ [5, thm:gcdle2]. Note this <i>extends</i> [4]’s own proved range, which was $e \leq 2$: e.g. $(e, f) = (4, 6)$ has $e > 2$ but $\gcd = 2$.
lem:const	8	Conditional — stated “granting” the previous line. Valid exactly for $\gcd(e, f) \leq 2$.
lem:char	8	Conditional , likewise.
prop:e1	8	Unconditional: a cyclic-shift argument on the one-bead relation, never mentioning nonvanishing. Reproved by [5, cor:coprime].
prop:suff	7	Unconditional and still true: the γ -family does commute. This is the sufficiency half of thm:main there and survives intact — it is the nowhere-zero part of Theorem 3.11, which [5, thm:main0] also proves.
thm:main, necessity	5	Valid exactly for $\gcd(e, f) \leq 2$. Superseded by [5, thm:main0], which reaches the same conclusion for $\gcd \leq 2$ and the correct larger one for $\gcd \geq 3$.
cor:size	5	True as stated about the group of admissible γ , but its <i>title</i> (“the size of the locus”) is wrong for $d \geq 3$: it sizes the γ -torus, which is then only the nowhere-zero part.

(P3) **“At least three quarters”, not “more than”.** The display of [6, cor:quarter] is $\#\text{supp}(\tau) \leq 2^{d-3}$, which is correct and *attained*: $\mathbb{K}_{\mathcal{C}_{j,b}}$ has support exactly 2^{d-3} and satisfies $(2B_d^*)$ (checked for all j, b and $3 \leq d \leq 8$). The prose gloss of that corollary therefore overstates its own display. Corollary 3.9 states the sharp form and records the attaining family.

(P4) **The $t^f - t^{f-2}$ witness is two-bead, not one-bead.** [2, prop:psiwitness] derives it from $t^{P+Q}(t^{-\varkappa} - t^{\varkappa})$ with $P = 0$, $Q = f - 1$, $\varkappa = -1$. The one-bead element of that paper is $(m(b + e) - m(b + f))(1 + t)t^{N-1}$. The note in §4 (§5) states this at the point of use.

6 Verification of the counts

Four numerical claims are made above: the border-split counts of Theorem 3.12 and Corollary 3.10, the antipalindrome and orbit counts of Theorem 3.13, the cylinder count $2\lfloor (d-1)/2 \rfloor$ and size 2^{d-3} of Definition 3.6, and the family/name discrepancy of Remark 2.7. All were re-enumerated on 18 September 2026 *from Convention 2.6 and Convention 2.3 as printed in this paper*, by code sharing nothing with the sources’ enumerators. This is the check that matters for (P1): the sources state these counts under a different name for the condition, so agreement of the numbers is what certifies that the condition imported here is the condition they counted.

d	border-split	antipalindromes	T -orbits	unbordered
2	0	0	0	2
3	2	2	2	2
4	4	0	2	4
5	8	4	6	6
6	12	0	6	12
7	28	8	18	20
8	48	0	24	40
9	104	16	60	74

Every entry agrees with the sources except the $d = 2$ orbit count, corrected in Remark 3.14. The last column is the check behind Remark 2.7: the two families first separate at $d = 5, 8$ against 6, the difference being exactly 0101 and 1010. The orbit column is additionally constrained by Burnside, $\# \text{orbits} = \frac{1}{2}(\# \text{words} + \# \text{fixed points})$, which the first two columns determine independently; it holds for $3 \leq d \leq 9$ and fails for the reported $d = 2$ value. That identity is what makes the correction decisive rather than a disagreement between two enumerations.

Remark 6.1 (no external base was found). It would be better to check these counts against a sequence this project did not generate. Two candidates were on hand: OEIS A299699 (rim-hook / border-strip tableaux of size n : 1, 4, 13, 50, 179, 744, 2975, ...) and A296561 (the same refined by shape, Heinz indexed). **Neither is the right base, and no substitute was sought.** They count border-strip *tableaux* — chains of ribbon additions filling a partition — whereas the counts above enumerate *binary words of length $d - 1$* subject to a border condition; the objects are different, the index means a different thing in each, and the growth rates differ ($\sim 4.5^n$ against $\sim 2^n$). No alignment or shift brings them together, and looking for a sequence that did match would produce a base chosen for matching, which is worth nothing. The counts above therefore rest on the internal Burnside consistency and on independent re-enumeration, and *not* on an external source. This is recorded as a limitation, not as a verification.

Remark 6.2 (locality: the intersection with Murnaghan–Nakayama). There is a natural class of weights that the present classification almost entirely misses, and saying which is worthwhile because it delimits the result. Korff’s Hecke rim-hook weight [8, eq. (Hecke_wt)], normalised as he normalises it ($\text{prob}(b) = (t - 1) \text{wt}(b)$), is in the coordinates of Convention 2.2

$$\sigma_{\text{Korff}}(u) = (t - 1) t^{e-1-|u|} = (t - 1) t^{e-1} (t^{-1})^{|u|},$$

using $\text{ht} + \text{wd} - 1 = e$ to trade the column count for the row count. So it is a *height* weight of geometric type $AB^{|u|}$: every bead the move jumps contributes the same factor $B = t^{-1}$, independently of *which* beads they are. Any weight assembled multiplicatively from per-letter data has this form. By Corollary 3.16, $[R_e^W, R_f^{\bar{W}}] = 0$ with both weights of this form forces $B = 1$: **the weights that are products over single letters and the weights that commute across ranks meet exactly in Murnaghan–Nakayama.** In particular no operator of Theorem 3.12 is of this form, for $\mathbb{K}_w$ is supported on one word and $AB^{|u|}$ is supported everywhere or nowhere.

This is the sense in which the product form of Theorem 3.11 is *not* a locality statement, despite looking like one: $\tau^{\otimes(e/d)}$ is a product over blocks of length d which *discards* every d -th letter, and it reduces to a product over single letters only when $d = 1$ — which is precisely Corollary 3.15.

Source caveat. The formula for σ_{Korff} above is verified directly against [8] (deep-read; and the operator form there, **lem:E**, contains a typographical inversion in its statement — his own proof, his definitions of $\psi^\pm(t)$, and his consistency condition $\text{prob} = (t - 1) \text{wt}$ all give $E_{ij}(t) = (1 - t^{-1})(-t)^{1-r}$, not the printed $(t^{-1} - 1)(-t)^{r-1}$). The further identification of the class $AB^{|u|}$ with “realisable

by a local six-vertex Boltzmann weight” rests on [9, Rmk. 2.9], which this project knows only at agent-summary depth. **That identification is therefore not claimed here as proved**; the statement above is about the class $AB^{|u|}$, which is defined internally, and only the name “local” is borrowed.

7 What is not proved

- (H1) **The exact rank- d form of $(2B_e^*)$ when $e > d$.** Theorem 3.11 imposes $(2B_e^*)$ on $\sigma = c\tau^{\otimes(e/d)}$, which is what the commutator hands over; but which *seeds* τ have $\tau^{\otimes k}$ satisfying $(2B_{kd}^*)$ is not known in closed form. What is known [5, rem:gap]: $(2B_d^{**}) \Rightarrow (2B_e^*) \Rightarrow (2B_d^*)$, and *both* implications are strict for $d \geq 4$, so the middle condition is genuinely new. Whether it depends on k is unknown; it does not for $k \in \{2, 3\}$ at $d \leq 4$. Theorem 3.12 is the first clean fact about it from the other side: for even d *no* single-shape τ satisfies $(2B_d^{**})$ — there are no antipalindromes of odd length — yet single-shape τ at $d = 4, 6, 8$ do give commuting pairs. That is exactly why those examples are new, and it is also why the $(2B^{**})$ route of [5, thm:counter] cannot reach them. Note this does not weaken any theorem above; it is a question about a *sufficient* condition, not about the classification. Theorem 3.13 adds one constraint the answer must satisfy: the middle condition is T -stable, since $(2B^*)$ is and $\otimes$ commutes with T .
- (H2a) **The interior of a cylinder is not described.** Corollary 3.9 bounds an admissible support by a cylinder and [6, cor:maximal] shows the cylinders are the maximal proper ones, but the admissible subsets *inside* a cylinder are not classified. The counts are 1, 3, 11, 31, 356. The obvious induction is **closed**: there is no uniform maximal admissible set one level down — at $m = 4$ the maximal proper admissible subsets of a cylinder have sizes $\{4, 4, 5, 5, 5, 5, 6, 6, 7, 9, 13\}$, so no single object plays the role the cylinders play one level up. [6, thm:closure] gives a generation rule but not a normal form.
- (H2b) **No uniform description of the value lattice L_Z .** Given an admissible support Z , the group of possible value-vectors is not described uniformly, and the sporadic 2-torsion in it is unexplained.
- (H2c) **The T -fixed locus is known only at the two extremes.** Theorem 3.13 determines the action of T on the nowhere-zero seeds (a scalar) and on the single-shape seeds (the free involution $w \mapsto w^T$). For a seed with zeros that is not a single shape, whether $\tau \circ T \in F^\times \tau$ is open.
- (H2d) **Reversal alone.** [7, prop:norev] rules out invertible *degree-preserving* implementations of rev; a degree-mixing one is not ruled out. Since rev acts nontrivially on the locus (it inverts γ), an implementation would be worth more than $\mathcal{P}$ is.

A Label $\rightarrow$ number map

Generated mechanically from the sources’ .aux files on 18 September 2026, not written by hand; regenerate before quoting any printed number, since the counters shift whenever a source is edited. Every row is a label this paper mentions. Q91, Q92 and Q147 use a flat theorem counter; Q149, Q150c1, Q150c2 and Q151 number by section. The 34 numbers carried over from the 17 September table were re-checked against the same .aux files and all 34 agreed. **Cite labels, not**

numbers: the numbers below are for a reader holding the PDFs, and they are the only numbers this paper asserts about its sources.

file	label	no.	what it is
Q91	lem:abacus	4	ribbon = bead move
Q91	lem:sign	5	the Jordan–Wigner sign
Q91	lem:dressing	6	the dressing factor
Q92	thm:matrix	5	the two sectors, single t
Q92	prop:psiwitness	16	the $t^f - t^{f-2}$ witness (two -bead, P4)
Q92	cor:three	19	—
Q147	lem:realise	4	every window occurs
Q147	lem:onebead	5	one-bead element, height form
Q147	lem:B	8	the forcing relation
Q147	lem:sectors	11	—
Q147	lem:twobead	12	—
Q149	lem:dict	1.3	word $\leftrightarrow$ composition
Q149	lem:transpose	1.6	transposition = rev-and-cpl
Q149	lem:realise	3.1	every window configuration occurs
Q149	lem:sectors	4.1	the two sectors, shape form
Q149	lem:onebead	4.2	one-bead element, shape form
Q149	lem:twobead	4.3	two-bead element, shape form
Q149	cor:size	5.4	size of the γ -torus (P2)
Q149	prop:suff	7.1	sufficiency; unconditional (P2)
Q149	lem:flip	8.1	(F0)–(F3); unconditional (P2)
Q149	lem:nonvanish	8.2	false for $\gcd \geq 3$ (P2)
Q149	lem:const	8.3	—
Q149	lem:char	8.4	—
Q149	prop:e1	8.5	—
Q150c1	thm:structure0	1.2	Lyndon–Schützenberger for $\otimes$
Q150c1	thm:main0	1.3	the classification (biconditional)
Q150c1	cor:main0	1.4	corrected nonvanishing
Q150c1	conv:2Bstar	2.1	$(2B^*)$, $(2B^{**})$
Q150c1	prop:1B	3.1	one-bead = $\otimes$ -commutation
Q150c1	prop:2B	3.3	two-bead = $(2B^*)$
Q150c1	thm:gcdle2	6.1	nonvanishing for $\gcd \leq 2$
Q150c2	def:S	2.2	admissible support
Q150c2	def:border	4.1	(BS), there called “non-overlapping” (P1)
Q150c2	thm:singleton	4.2	$\nVdash_w$ is $(2B^*)$ iff (BS)
Q150c2	rem:endrows	4.3	the end rows
Q150c2	thm:singleton2	4.5	$(2B^{**})$ iff antipalindrome
Q150c2	cor:48	5.2	the smallest new case, $d = 4$
Q150c2	cor:three	6.2	the $\gcd \leq 2$ case, reproved
Q150c2	rem:sym	6.6	the word-level $(\mathbb{Z}/2)^2$
Q151	lem:Ctrans	1.4	$\mathcal{P}$ on occupancy words
Q151	thm:A	2.1	$\mathcal{P}R_e^W\mathcal{P} = R_e^{W\circ T}$
Q151	thm:C	4.4	not Terwilliger’s $\dagger$
Q151	thm:D	4.5	action on the locus

file	label	no.	what it is
Q151	cor:parity	4.6	the parity count
Q151	prop:norev	4.7	no implementation of rev alone
Q151	cor:accounting	4.9	the $(\mathbb{Z}/2)^2$ accounting

References

- [1] Clio, *The fermionic normal form of the ribbon operator*, proofs/2026-09-06-c2-Q91-fermionic-normal-form.tex, 6 September 2026.
- [2] Clio, *The cross-rank ribbon commutator*, proofs/2026-09-07-Q92-cross-rank-commutator.tex, 7 September 2026.
- [3] Clio, *Free height weights force Murnaghan–Nakayama*, proofs/2026-09-11-c2-Q147-multi-t-bracket.tex, 11 September 2026.
- [4] Clio, *Shape weights on ribbons*, proofs/2026-09-15-c1-Q149-shape-weights.tex, 15 September 2026.
- [5] Clio, *Q150: the nonvanishing lemma is false*, proofs/2026-09-16-c1-Q150-nonvanishing.tex, 16 September 2026.
- [6] Clio, *Q150 (H2): single-shape commuting operators*, proofs/2026-09-16-c2-Q150-H2-single-shape.tex, 16 September 2026.
- [7] Clio, *Particle–hole conjugation is not Terwilliger’s $\dagger$* , proofs/2026-09-17-c1-Q151-particle-hole.tex, 17 September 2026.
- [8] C. Korff, *Cylindric Hecke characters and Gromov–Witten invariants via the asymmetric six-vertex model*, arXiv:1906.02565.
- [9] L. Petrov, *(Mittag-Leffler lecture notes)*, arXiv:2609.18502. Known to this project at agent-summary depth only; cited once, in Remark 6.2, for a statement explicitly not claimed as proved.
- [10] Clio, working memoranda, memory/MEMORY.md.